

Customer Shopping Behavior Analysis Report

Project Overview

This project demonstrates an end-to-end data analytics workflow using Python, MySQL, and Power BI. The raw customer shopping dataset was first cleaned and transformed in Jupyter Notebook using Python. The cleaned dataset was then loaded into MySQL, where SQL queries were used to analyze customer behavior, sales performance, purchasing patterns, and other business metrics. Finally, the insights were visualized through an interactive Power BI dashboard.

Business Problem Statement

Retail businesses collect large amounts of customer transaction data, but raw data alone does not provide useful business insights. The objective of this project is to convert raw shopping data into meaningful information that helps understand customer behavior, product performance, sales trends, and revenue generation. These insights can support better business decisions and improve customer satisfaction.

Exploratory Data Analysis (EDA)

The dataset was explored to understand its structure, identify missing values, detect inconsistencies, and discover trends. The EDA process included:

- Understanding data types and column distributions.
- Handling missing or inconsistent values.
- Analyzing customer demographics.
- Studying purchase amounts across different age groups.
- Examining category-wise sales and revenue.
- Evaluating subscription status and customer engagement.
- Reviewing customer ratings and purchasing patterns.

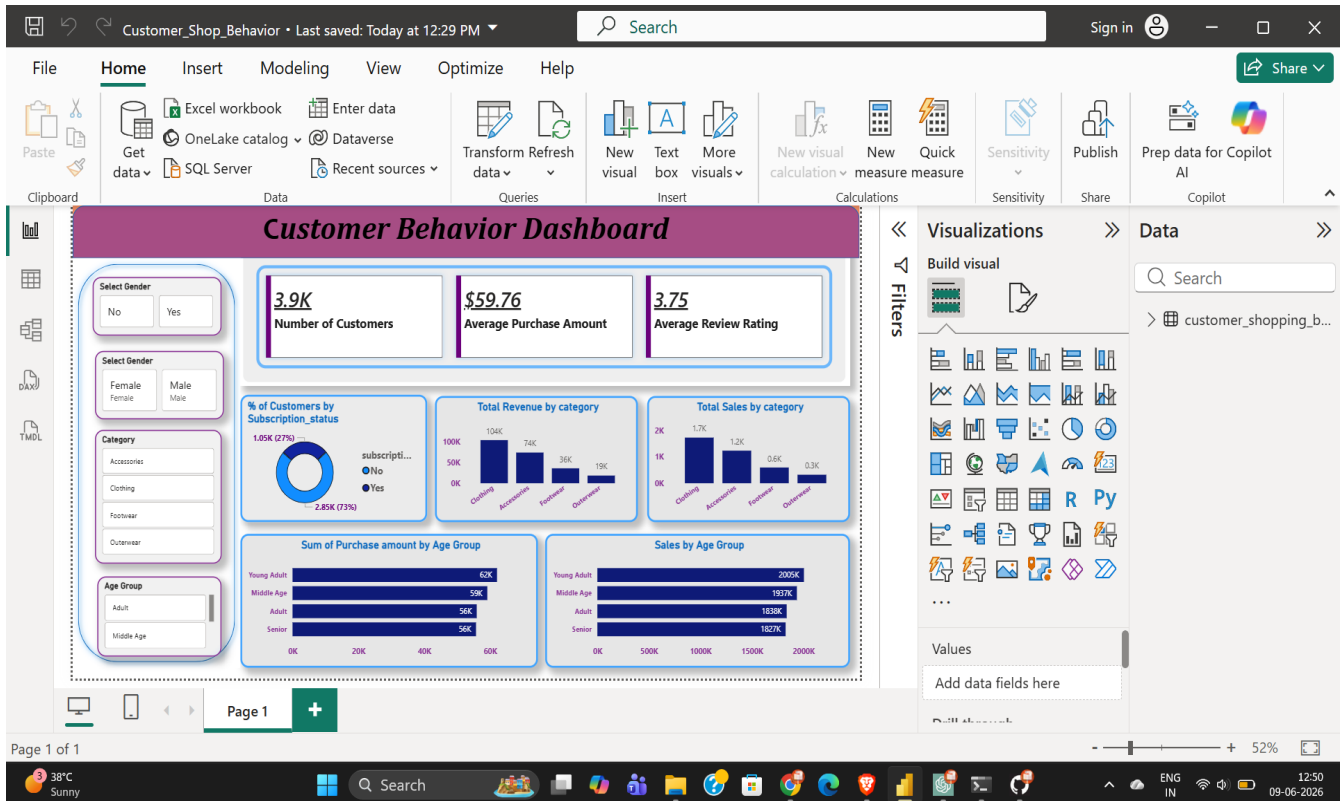
Tools and Technologies

Python (Pandas, NumPy), Jupyter Notebook, MySQL, SQL, and Power BI.

Project Deliverables

- Cleaned and transformed dataset.
- Jupyter Notebook for preprocessing and data cleaning.
- MySQL database containing processed data.
- SQL queries used for business analysis.
- Business insights derived from SQL analysis.
- Interactive Power BI dashboard.
- Final project report and documentation.

Power BI Dashboard



Dashboard Insights

- Customer count is displayed through KPI cards.
- Average purchase amount and review ratings are highlighted.
- Category-wise revenue and sales performance are visualized.
- Subscription status distribution helps understand customer loyalty.
- Age-group analysis identifies key customer segments.
- Interactive filters allow detailed exploration of customer behavior.

Conclusion

The project successfully transformed raw customer shopping data into actionable business insights. Using Python for data preparation, MySQL for analysis, and Power BI for visualization, the project demonstrates a complete data analytics workflow that can support data-driven decision-making.